5.9 MBPD*
Expected global oil demand by 2028
*Million barrels per day

10%
Share of biofuels in liquid fuel supply growth by 2028

1,016
CRORE LITRES PER ANNUM
India's ethanol market in the next two years

31%
Expected CAGR of India's ethanol market between 2023 & 2026

THE CHANGE-MAKERS

TRUALT BIOENERGY

- With an installed capacity of 2,000 kilolitres per day (KLD) in its three plants in Karnataka's Bagalkot district, the Bengaluru-based firm is one of the largest contributors to the ethanol blending programme
- The company has access to a diverse portfolio of energy products—bagasse to produce second generation (2G) ethanol in the future and state-of-the-art infrastructure to make multiple upstream products such as sustainable aviation fuel, methane and bio CNG in a very short time
- The current focus is on capacity expansion to 5,000 KLD, introducing multi-feed facilities to utilise all possible raw materials, the production of various upstream products such as sustainable aviation fuel and molecular grade ethanol
- The plan is to set up 250 biofuel dispensing stations—the first one has already come up in Bagalkot

EMAMI AGROTECH

- The Kolkata-based company is the largest biodiesel manufacturer in the eastern region and among the top three producers of biodiesel in India
- Its multi-feed stock plant in Haldia, West Bengal—set up in 2005 with technical knowhow from Belgium's Desmet Ballestra—produces 350 tonnes a day
- The company sells its product to various oil marketing companies, the railways, transport sector and mining industry either as B100 or in blended form with diesel

Other biofuels include methane gas and biogas.

The Centre's Ethanol Blended Petrol programme under the National Policy on Biofuels, 2018, laid out an indicative target of 20 per cent blending of ethanol in petrol by 2030. The target year was later advanced to 2025-26 in the May 2022 amendments to the policy that allows for more feed-stock for biofuel production. It also seeks to promote biofuel production under the Make in India programme at Special Economic Zones and Export Oriented Units.

With a compounded annual growth rate of 31 per cent between 2023 and 2026, the ethanol market in India is set to drastically reduce the country's dependence on imported crude oil, say industry sources. India's ethanol market is poised to grow to a minimum requirement of 1,016 crore litres per annum in the next two years. This will be primarily driven by increased ethanol blending ratio, higher fuel consumption, more vehicles, and multiple upstream applications of ethanol, biofuels and byproducts.



"We wanted to be part of the biodiesel industry whenever it opened up. Biodiesel is still very small in India though it is growing in Europe and the US. Availability of raw materials is most important"

–ADITYA V. AGARWAL, *Director, Emami Group*

WHAT INDIA NEEDS TO MASTER

India's transition effort comes with its own share of challenges too. The issues flagged by agriculture experts include the absence of a policy mechanism that incentivises farmers to collect and deliver biomass residues to a next-generation ethanol plant, high cultivation cost, low productivity, improper availability of seeds and unsuitable marketing channels.

If addressed efficiently and swiftly, India's journey towards a greener future will be that much faster. ■